Visualizing the Mandelbrot Set Boundary

A Python renderer exploring escape-time coloring and boundary zoom regions of the Mandelbrot set.

The math

A point c is in the Mandelbrot set if the orbit of 0 under $z_{n+1} = z_n^2 + c$ stays bounded. If $|z_n| > 2$ for any n, the orbit is guaranteed to escape.

What's in the repo

mandelbrot.py -- vectorized escape-time computation with numpy. render.py -- smooth vs. discrete coloring schemes.

Status

Working renderer; next step is arbitrary-precision arithmetic to push zoom depth further.